

Zero private activation beyond antidegradability

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Abstract

A noisy two-qubit Pauli channel is transpose-antidegradable and completely anti-less-noisy but not antidegradable. Exact rational signed-lift certificates and a sharp symmetric-extension threshold provide a robust counterexample to a proposed antidegradable-helper activation implication.

1 The result

Antidegradability means that a channel's output can be reconstructed physically from its environment. Transpose-antidegradability requires reconstruction only after output transposition. A recent question asks whether transpose-antidegradability together with complete environmental information dominance already forces antidegradability [2, Eq. (6.1)]. The following explicit family answers this negatively.

Let X, Y, Z be the usual Pauli matrices, and juxtapose their symbols for tensor products. Set

$$\mathcal{S} = \{XX, YI, IZ, XZ, IY, YZ\}, \quad \mathcal{F}(A) = \frac{1}{6} \sum_{P \in \mathcal{S}} PAP, \quad \mathcal{R}(A) = \text{Tr}(A) \frac{\mathbb{I}_4}{4},$$

and $\mathcal{N}_q = (1 - q)\mathcal{F} + q\mathcal{R}$. The channel $\mathcal{F}$ is the six-Pauli construction of [1]. The calculation here adds a noise interval on which its activation behavior changes before it becomes antidegradable.

Theorem 1. *For $0 \leq q \leq 1$, the channel $\mathcal{N}_q$ is transpose-antidegradable. It admits a complete variance-dominating Hermiticity-preserving signed lift through its complement whenever $q \geq 1/10$. It is antidegradable exactly when*

$$q \geq q_{\text{AD}} := \frac{3 - \sqrt{5}}{5}.$$

Consequently, for every $q \in [1/10, q_{\text{AD}})$, it is transpose-antidegradable and not antidegradable, but its complement dominates it in unrestricted complete relative-entropy order. Its private and quantum capacities remain zero after tensoring with any antidegradable channel.

Here “complete” quantifies every finite untouched reference and all state pairs. The proof needs only the established sufficient signed-lift criterion [2]; it does not assume necessity of that criterion. The complete-order threshold below $1/10$ is not determined.

2 Notation and the transpose simulator

Index the sixteen Hermitian Pauli matrices by $a = (x_1, z_1, x_2, z_2) \in \mathbb{F}_2^4$, in lexicographic order, using

$$P_{00} = I, \quad P_{10} = X, \quad P_{01} = Z, \quad P_{11} = Y$$

on each qubit. The active integer labels are

$$\mathbf{A} = \{1, 3, 9, 10, 12, 13\}.$$

Addition of labels means XOR. Define

$$\begin{aligned}\chi(a, g) &= (-1)^{x_1(a)z_1(g)+z_1(a)x_1(g)+x_2(a)z_2(g)+z_2(a)x_2(g)}, \\ \kappa(a, g) &= \frac{1}{4} \text{Tr}(P_a P_g P_{a+g}).\end{aligned}$$

Thus $P_a P_g = \kappa(a, g) P_{a+g}$, and $\kappa(a+g, g) = \overline{\kappa(a, g)}$.

First take $q = 1/10$ and abbreviate $\mathcal{N} = \mathcal{N}_{1/10}$. Put $r_a = 5$ for $a \in \mathbf{A}$ and $r_a = 1$ otherwise. Then

$$p_a = \frac{r_a^2}{160}, \quad V = \frac{1}{\sqrt{160}} \sum_a r_a P_a \otimes |a\rangle, \quad \mathcal{E} = \mathcal{N}^c.$$

In particular, $\mathcal{E}(A)_{ab} = \sqrt{p_a p_b} \text{Tr}(P_b P_a A)$.

On the active six-dimensional space, let

$$W|P\rangle = (I \otimes P)|\Phi_4\rangle, \quad |\Phi_4\rangle = \frac{1}{2} \sum_{j=1}^4 |jj\rangle.$$

The Bell vectors are orthonormal, so tracing the second output of W gives a CPTP map

$$\mathcal{D}_0(|P\rangle\langle Q|) = \frac{(QP)^T}{4}.$$

Pauli-basis substitution gives $\mathcal{D}_0 \mathcal{F}^c = T\mathcal{F}$. Define on the minimal environment of $\mathcal{N}$

$$\mathcal{D}(|a\rangle\langle b|) = \frac{24}{25} \mathbf{1}_{\mathbf{A}}(a) \mathbf{1}_{\mathbf{A}}(b) \frac{(P_b P_a)^T}{4} + \delta_{ab} n_a^2 \frac{I_4}{4}, \quad (1)$$

where $n_a^2 = 1/25$ on $\mathbf{A}$ and $n_a^2 = 1$ off it. The first summand is a compressed CP map and the second is a nonnegative measure-and-prepare map. Diagonal traces add to one and off-diagonal traces vanish. Thus $\mathcal{D}$ is CPTP. Its active-block coefficient is $(24/25)(25/160) = (9/10)/6$, and $\sum_a p_a n_a^2 = 1/10$. Therefore

$$\mathcal{D}\mathcal{E} = \frac{9}{10} T\mathcal{F} + \frac{1}{10} \mathcal{R} = T\mathcal{N}.$$

For general $q > 0$, the active and inactive probabilities are $(8 - 5q)/48$ and $q/16$. In (1), replace $24/25$ by $8(1 - q)/(8 - 5q)$ and use the complementary diagonal weights on the active coordinates. The same calculation proves the first assertion of Theorem 1; at $q = 0$ use $\mathcal{D}_0$.

3 An exact complete signed lift

The Pauli multipliers of $\mathcal{N}$ are $\lambda_0 = 1$ and $\lambda_g = 3s_g/10$ for $g \neq 0$, where

$$(s_1, \dots, s_{15}) = (+, -, -, -, -, -, +, +, -, -, -, +, +, -, +).$$

They can equivalently be obtained from $\lambda_g = \sum_a p_a \chi(a, g)$.

For $g \neq 0$ and $b = a + g$, define the real coefficient f_{ag} by the following five-value rule:

$$\frac{f_{ag}}{s_g} = \begin{cases} 25/2, & a, b \in \mathbf{A}, \\ 13/2, & \text{exactly one of } a, b \in \mathbf{A}, \chi(a, g) = s_g, \\ 5/4, & \text{exactly one of } a, b \in \mathbf{A}, \chi(a, g) = -s_g, \\ -9/2, & a, b \notin \mathbf{A}, \chi(a, g) = s_g, \\ 1/4, & a, b \notin \mathbf{A}, \chi(a, g) = -s_g. \end{cases}$$

Set $\mathcal{J}(I_4) = I_{16}$ and

$$\mathcal{J}(P_g)_{ab} = \begin{cases} \frac{\kappa(a, g)f_{ag}}{r_a r_b}, & b = a + g, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and extend complex linearly. All coefficients are Gaussian rational. Since $f_{ag} = f_{a+g, g}$, the matrices $\mathcal{J}(P_g)$ are Hermitian. Hence $\mathcal{J}$ is HP and unital.

Reconstruction. Counting the sixteen labels gives $\sum_a f_{ag} = 48s_g$. The definitions imply

$$(I \otimes \mathcal{J}(P_g))V = \frac{1}{\sqrt{160}} \sum_a \frac{f_{ag}}{r_a} P_a P_g \otimes |a\rangle.$$

Consequently

$$\mathcal{E}^* \mathcal{J}(P_g) = \frac{1}{160} \left(\sum_a f_{ag} \right) P_g = \lambda_g P_g = \mathcal{N}^*(P_g).$$

The identity direction follows from unitality, so $\mathcal{E}^* \mathcal{J} = \mathcal{N}^*$.

The full Gram certificate. Let $H_{ag} = \chi(a, g)$ for $g = 1, \dots, 15$, and set

$$G = \frac{1}{160} [H^T \text{diag}(r_a^2) H - f^T \text{diag}(r_a^{-2}) f]. \quad (3)$$

This explicit rational symmetric matrix has spectrum

Eigenvalue	Multiplicity
19/400	1
39/640	4
$(1877 - 4\sqrt{38119})/16000$	5
$(1877 + 4\sqrt{38119})/16000$	5

For a reproducible rational identity, its characteristic polynomial is

$$\det(xI - G) = \left(x - \frac{19}{400}\right) \left(x - \frac{39}{640}\right)^4 \left(x^2 - \frac{1877}{8000}x + \frac{116529}{10240000}\right)^5. \quad (4)$$

Equations (3)–(4) are checked by finite rational arithmetic from the five-value rule. All roots are positive; for the smaller quadratic root, $1877^2 - 16 \cdot 38119 = 2913225 > 0$.

For $g, h \neq 0$, define

$$\Gamma_{gh} = \mathcal{N}^*(P_g P_h) - \mathcal{E}^*(\mathcal{J}(P_g) \mathcal{J}(P_h)).$$

Writing $N_g = (P_g \otimes I)V$ and $K_g = (I \otimes \mathcal{J}(P_g))V$, direct substitution gives

$$\Gamma_{gh} = N_g^* N_h - K_g^* K_h, \quad P_g \Gamma_{gh} P_h = G_{gh} I_4.$$

Thus the 60×60 Gram matrix $[\Gamma_{gh}]$ is unitarily congruent to $G \otimes I_4$. The identity row and column vanish by reconstruction. The entire Gram defect is positive semidefinite. Expanding any matrix-amplified observable in the Pauli basis therefore proves

$$(\mathcal{E}^* \otimes \text{id})(\mathcal{J}(X)^* \mathcal{J}(X)) \preceq (\mathcal{N}^* \otimes \text{id})(X^* X) \quad (5)$$

at every reference dimension. This verifies genuine complete square domination, including HP, rather than a complex-left relaxation.

For a Hermitian receiver score, use its Hermitian image under $\mathcal{J}$ in the variational formula for the SLD quadratic form. Reconstruction preserves the linear term and (5) decreases the quadratic penalty. Hence $\mathcal{E}$ completely SLD-dominates $\mathcal{N}$. The established coherent-reference implication gives complete relative-entropy and less-noisy dominance [2].

For $q \geq 1/10$, write $\mathcal{N}_q = t\mathcal{N}_{1/10} + (1-t)\mathcal{R}$, with $t = 10(1-q)/9$. In a Kraus environment retaining the mixture flag, combine the two lifts block-diagonally. Reconstruction and square domination are their convex combinations. Complementary channels related by an isometry give the same comparison. This proves the second assertion of Theorem 1.

4 The sharp antidegradability threshold

Let

$$\Pi = \sum_{P \in \mathcal{S}} |P\rangle_{\text{Bell}} \langle P|, \quad |P\rangle_{\text{Bell}} = (I \otimes P)|\Phi_4\rangle.$$

The normalized Choi state is

$$\rho_q = \frac{1-q}{6}\Pi + \frac{q}{16}I_{16}.$$

Antidegradability is equivalent to a positive normalized extension on AB_1B_2 having both AB_i marginals equal to ρ_q .

Define $A = \Pi_{AB_1} + \Pi_{AB_2}$, with identities on the unused factors, and $M = 4A$. In the computational basis, M is an integer matrix and obeys the exact identity

$$M[(M - 5I)^2 - 5I][(M - 3I)^2 - 5I] = 0. \quad (6)$$

Because M is Hermitian, (6) implies $\|A\| \leq (5 + \sqrt{5})/4$. Every extension would therefore satisfy

$$2 \operatorname{Tr}(\Pi \rho_q) = 2 \left(1 - \frac{5q}{8}\right) \leq \frac{5 + \sqrt{5}}{4}.$$

This forces $q \geq (3 - \sqrt{5})/5$. At $q = 1/10$, the strict separating gap is $(5 - 2\sqrt{5})/8$.

The bound is attained. With $s = \sqrt{5}$, put

$$P_{\max} = \frac{M[(M - 3I)^2 - 5I][M - (5 - s)I]}{80(3 + s)}. \quad (7)$$

Equation (6) makes this the spectral projection at $5 + s$. Direct substitution gives

$$\operatorname{Tr} P_{\max} = 12, \quad \operatorname{Tr}_{B_2}(P_{\max}/12) = \rho_{(3-s)/5}.$$

It is invariant under exchanging B_1, B_2 , so the other marginal agrees. Thus $P_{\max}/12$ is the required positive symmetric extension at the boundary. Convex mixing with the replacer extends antidegradability to all larger q , completing the proof.

5 Operational conclusion and limitations

Complete environmental relative-entropy dominance tensorizes. An antidegradable helper has the same dominance property by data processing. Their product therefore has zero private capacity

and, consequently, zero quantum capacity. This proves the claimed failure of antidegradable-helper activation throughout the nonempty interval

$$\boxed{\frac{1}{10} \leq q < \frac{3 - \sqrt{5}}{5}}.$$

The source’s proposed implication is therefore false. This counterexample does not determine a general complete-order membership algorithm, establish necessity of signed lifts, or locate the family’s complete-order transition below $q = 1/10$. The sharp threshold proved here concerns antidegradability. All members of the family already have zero unassisted private capacity by transpose simulation.

The displayed identities have been checked independently in exact arithmetic. The result is a mathematical research note, not a claim of external peer review or exhaustive historical priority.

References

- [1] Uthirakalyani G, P. Halder, and D. Elkouss. *Private communication from Pauli channels with no privacy*. arXiv:2609.20133 (2026).
- [2] C. Hirche. *Quantum estimation, channel orders, and private capacity*. arXiv:2610.03706v1 (2026). In particular, the sufficient complete signed-lift criterion, tensorization, and the question in Eq. (6.1).